

Stereo-rig Jetson Setup

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1. Install JetPack

- 1) Prepare Ubuntu 20.04 machine
- 2) Download Nvidia SDK(<https://developer.nvidia.com/sdk-manager#getting-started>)
- 3) Connect Jetson pins 9 (GND) and 10 (REC) to enter recovery mode. The pins are located under the heat sink.
- 4) Connect the Jetson to your machine using a USB-C to USB-A cable.
- 5) Power on the Jetson and install Jetpack(refer this video: <https://www.youtube.com/watch?v=CmDaXah4l5g>)

2. Install Docker

Shell

```
sudo apt update
sudo apt install -y ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
```

```

Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update

# Install Docker
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin

# Start Docker
sudo systemctl enable --now docker

# Test
sudo docker run hello-world

```

3. Time Synchronization

a. Primary

* Please check if the RTC module is correctly connected.*

Set RTC time as default

```

Shell

# Find the RTC device
ls -l /dev/rtc*
for r in /sys/class/rtc/rtc*; do
echo "$r: $(cat $r/name)"
done

## Set the RTC time(Need internet connection)
sudo hwclock --show --utc --rtc /dev/rtc0
sudo hwclock --show --utc --rtc /dev/rtc2
date -u

# Copy /dev/rtc0 time into /dev/rtc2

```

```

sudo hwclock --hctosys --utc --rtc /dev/rtc0
sudo hwclock --systohc --utc --rtc /dev/rtc2

# Check
sudo hwclock --show --utc --rtc /dev/rtc0
sudo hwclock --show --utc --rtc /dev/rtc2
date -u

# Use /dev/rtc2 as time source at boot
sudo nano /etc/systemd/system/rtc-to-system.service

...

[Unit]
Description=Set system clock from RTC module before chrony
DefaultDependencies=no
Wants=systemd-udev-settle.service
After=systemd-udev-settle.service
Before=chrony.service time-sync.target

[Service]
Type=oneshot
ExecStart=/usr/sbin/hwclock --hctosys --utc --rtc /dev/rtc2

[Install]
WantedBy=sysinit.target
...

# Reload and enable
sudo systemctl daemon-reload
sudo systemctl enable rtc-to-system.service
sudo systemctl restart rtc-to-system.service

# Check
systemctl status rtc-to-system.service
date -u
sudo hwclock --show --utc --rtc /dev/rtc2

```

Install and set chrony

```

Shell
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.0.150/24 \
  ipv4.method manual \
  ipv4.never-default yes

```

```

# Restart the Ethernet connection
sudo nmcli con up "Wired connection 1"

# Check
ip route

# Install chrony
sudo apt update
sudo apt install chrony
sudo systemctl disable --now systemd-timesyncd
sudo systemctl enable chrony

# Edit
sudo nano /etc/chrony/chrony.conf

...

# Master Jetson chrony server

allow 192.168.0.0/24
bindaddress 192.168.0.150

# Serve local system time after it is initialized from RTC
local stratum 8

driftfile /var/lib/chrony/chrony.drift
makestep 1.0 3
...

# Save, then restart chrony
sudo systemctl restart chrony

# Check
systemctl status chrony
chronyc tracking

```

b. Secondary

```

Shell
sudo nmcli con mod "Wired connection 1" \
    ipv4.addresses 192.168.0.100/24 \

```

```
    ipv4.method manual \
    ipv4.never-default yes

# Restart
sudo nmcli con up "Wired connection 1"

# Check
ip route

# Test Ethernet communication
ping 192.168.0.150

# Install chrony
sudo apt update
sudo apt install chrony
sudo systemctl disable --now systemd-timesyncd
sudo systemctl enable chrony

# Add
sudo nano /etc/chrony/chrony.conf

...

# Client Jetson follows master Jetson

server 192.168.0.150 iburst prefer

driftfile /var/lib/chrony/chrony.drift
makestep 1.0 3
...

# Restart
sudo systemctl restart chrony
sudo chronyc makestep

# Check
chronyc sources -v
chronyc tracking
```